Mr. V. ARULALAN M.Tech., (Ph.D.),



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CAREER OBJECTIVE

To discharge my duties as an able academician and an effective researcher in a serene and sound environment of students, and to make them strongly competent to face the current world. I also wish to train the students technically so as to impart them ethics of life and mould their personality to be a dynamic Engineer in today's competitive world.

PROFESSIONAL EXPERIENCE

S. No.	DESIGNATION	INSTITUTION	PERIOD
1	Teaching Fellow	Department of Information Technology, MIT Campus, Anna University, Chennai – 600044	November 2020 – till date
2	Research Fellow	Department of Information Technology, MIT Campus, Anna University, Chennai – 600044	January 2018–October 2020
3	Assistant Professor	Department of Computer Science and Engineering, Christ College of Engineering & Technology, Puducherry – 605010	June 2016 –December 2017

EDUCATIONAL QUALIFICATION

DEGREE	INSTITUTION	BOARD / UNIVERSITY	YEAR OF PASSING	%/CGPA
Ph.D.	Madras Institute of Technology Campus, Chennai.	Anna University	***Thesis Submitted*** November 2022	
M.Tech. (CSE)	Manakula Vinayagar Institute of Technology, Puducherry	Pondicherry University	2015	8.31
B.Tech. (CSE)	R V S College of Engineering & Technology, Karaikal, Puducherry	Pondicherry University	2013	6.85
H.S.L.C.	Sri Sankara Vidyalaya Matriculation Higher Secondary School, Puducherry	Tamilnadu State Board	2008	64.66%
S.S.L.C.	Petit Seminaire Higher Secondary School, Puducherry	Tamilnadu Matriculation Board	2006	56.45%

SCHOLASTIC ACHIEVEMENTS

1. Received **Anna Centenary Research Fellowship (ACRF 2018)** scholarship from Anna University, INDIA.[Award Letter No.: CFR/ACRF/2018/AR1/35]
2. Awarded **Mentor Certificate** for the course "Introduction to Modern Application Development" from **NPTEL** organized by IIT-Madras in partnership with NASSCOM.

SKILL SET

- Programming Languages : Python, C, C++
- Software Skills : MATLAB, Apache Hadoop, Git
- Cloud Technologies : AWS and GCP
- Web Development : PhP, HTML, CSS, MySQL
- Operating Systems : Windows and Linux

AREAS OF INTEREST

- Video Processing and Analytics
- Machine & Deep Learning
- Data Structures
- Algorithm Analysis
- Biometric Security

SUBJECTS HANDLED

- IT5201 - Information Technology Essentials
- IT5603 - Distributed and Cloud Computing
- IT5501 - Web Technologies
- IT7016 - Programming with Open Source Software
- IT5039 - Deep Learning
- IT5302 - Software Engineering
- OE5091 - Business Data Analytics
- IT5412 - Advanced Data Structures Laboratory
- IT5411 - Operating System Laboratory
- IT5511 - Computer Networks Laboratory

TRAINING & CERTIFICATION

1. Completed an online course **"Deep Learning Specialization"** authorized by deeplearning.ai and offered through Coursera. Grade Achieved: 99.02%. Credential ID: *3QBY3JTQHUSX*

2. Completed an online course **"Data Engineering, Big Data, and Machine Learning on GCP Specialization"** authorized by Google Cloud and offered through Coursera. Grade Achieved: 98.74%. Credential ID: *8ZGA8DVL3B5T*

3. Completed an online course **"Object Detection with Amazon Sagemaker"** authorized by Amazon Web Services and offered through Coursera. Grade Achieved: 80%. Credential ID: TE2LDPMDFUVZ

4. Completed an online course **"Machine Learning for Business Professionals"** authorized by Google Cloud and offered through Coursera. Grade Achieved: 98.33%. Credential ID: **3**M54647LX8K4

5. Completed an online course **"Computer Vision Basics"** authorized by University at Buffalo, The State University of New York, and offered through Coursera. Grade Achieved: 100%. Credential ID: BHFK6VH6JFRL

6. Completed an online course **"Getting Started with AWS Machine Learning"** authorized by Amazon Web Services and offered through Coursera. Grade Achieved: 100%. Credential ID: KMFV486GU8JM

7. Completed an online course **"Building Conversational Experiences with Dialogflow"** authorized by Google Cloud and offered through Coursera. Grade Achieved: 100%. Credential ID: G35F69EK75JF

8. Completed an online course **"Introduction to Cloud Identity"** authorized by Google Cloud and offered through Coursera. Grade Achieved: 88.66%. Credential ID: HEAUEUVRFJRL

9. Completed 8 Weeks **NPTEL** course "**Introduction to Modern Application Development**" with a consolidated score of **70%** organized by IIT-Madras in partnership with NASSCOM. Roll No: NPTEL17CS0626640346FN

RESEARCH PROFILE

A. DOCTORAL RESEARCH

Deep Learning Based Methods for Improving Object Detection Classification and Tracking in Video Surveillance

Summary:

The research dealt with detection and tracking of objects using the Modified Manta-Ray Foraging Optimization-based Convolution Neural Network (M2RFO-CNN) in a complex dynamic environment. Also, an efficient object detection and tracking in adverse weather conditions is focused in this research.

Status: Thesis Submitted on November 2022**.**

Details of the Supervisor:

Dr.Dhananjay Kumar M.E., M.Tech., PhD.,

Professor and Head,
University Department of Information Technology,
Anna University, MIT Campus, Chennai – 600044.
dhananjay@annauniv.edu
Ph : 044 – 2251 6020 / 6021 / 6306
Mob : +91 9840598722

B. PATENT

IoT Based Novel One Touch Wearable Smart Bracelet for Stress Prediction

Registration Number	:**1193455**
Status	:Registered
Registered	:2022-05-24
Type	:Copyright
Agency	:Canadian Intellectual property and copyright

C. RESEARCH PUBLICATIONS

IN PEER-REVIEWED INTERNATIONAL JOURNALS

1. V. Arulalan and Dhananjay Kumar, "**Efficient object detection and classification approach using HTYOLOV4 AND M^{2}RFO-CNN**," Computer Systems Science and Engineering, vol. 44, no.2, pp. 1703–1717, 2023. [**Science Citation Index**]

2. **V. Arulalan**, G. Balamurugan, V. Premanand, "**Multi-Modal Biometric Recognition System using Palmprint and Inner-Knuckle Print**", International Journal of Applied Engineering Research, Volume 10, Number 14 (2015) pp 34748-34751. [**SCOPUS Indexed**]

3. **V. Arulalan**, V. Premanand, G. Balamurugan, "**An Overview on multi Modal Biometrics**", International Journal of Applied Engineering Research, Volume 10, Number 17 (2015) pp 37534-37538. [**SCOPUS Indexed**]

4. V. Premanand, A. Meiappane, **V. Arulalan**, "**Survey on Captcha and its Techniques for BOT Protection**", International Journal of Computer Applications (0975 – 8887) Volume 109 – No. 5, January 2015. [**SCOPUS Indexed**]

5. **V. Arulalan**, N. Geetha, V. Premanand, "**Multimodal Biometric System using Iris and Inner-Knuckle Print**", International Journal of Computer Applications (0975 – 8887) Volume 106 – No.6, November 2014. [**SCOPUS Indexed**]

6. G. Balamurugan, **V. Arulalan**, V. Lokesh, "**Customer - Management Relationship Service using Android**", International Journal of Computer Applications (0975 – 8887) Volume 105 – No. 10, November 2014. [**SCOPUS Indexed**]

7. G. Balamurugan, Dr. K. B. Jayarraman, **V. Arulalan**, "**A Survey on Medical Image Watermarking Techniques**", IJCSN International Journal of Computer Science and Network, Volume 3, Issue 5, October 2014.

8. G. Balamurugan, Dr. K. B. Jayarraman, **V. Arulalan**, "**Multi-Modal Biometric Template Protection Using Image Processing**", International Journal of Engineering Research & Technology (IJERT) Vol. 3 Issue 4, April – 2014.

9. **V. Arulalan**, G. Balamurugan, V. Premanand, "**A Survey on Biometric Recognition Techniques**", International Journal of Advanced Research in Computer and Communication Engineering Vol. 3, Issue 2, February 2014.

10. G. Balamurugan, Sakthivel Punniakodi, **V. Arulalan**, C. Jayanavithraa, "**Multipath Forwarding in VANET for Message Dissemination**", International Journal of Advanced Research in Computer and Communication Engineering Vol. 2, Issue 11, November 2013.

IN PEER-REVIEWED INTERNATIONAL CONFERENCES

11. **V. Arulalan**, Dhananjay Kumar, V. Premanand, "**An Item Based Collaborative Filtering For Similar Movie Search**", International Conference on Computational Vision and Bio-Inspired Computing-**SPRINGER**, 2018. [**SCOPUS Indexed**]

12. V. Premanand, Dhananjay Kumar, **V. Arulalan**, "**An Instance İdentification Using Randomized Ring Matching via Score Generation**", International Conference on Computational Vision and Bio-Inspired Computing-SPRINGER, 2018. [**SCOPUS Indexed**]

13. G. Balamurugan, **V. Arulalan**, K. Suresh Joseph, "**Digital Image Protection Using Reversible Watermarking and Private Symmetric Key Encryption Technique**", International Journal of Advanced Information Science and Technology (IJAIST), Vol.6, No.1, ISSN: 2319:2682, November 2017, 1st International Conference on Recent Trends in Engineering & Technology, ICRTET'17. [**SCOPUS Indexed**]

14. **V. Arulalan**, Dr. K. Suresh Joseph, V. Premanand, "**Securing Digital Data using 256-bit Multimodal Biometrics based Cryptographic Key**", **IEEE** Sponsored World Conference on Futuristic Trends in Research and Innovation for Social Welfare (**WCFTR'16**), ISBN: 978-1-4673-9214-3, March 2016.

15. G. Balamurugan, Dr. K. Suresh Joseph, **V. Arulalan**, "**An Iris based Reversible Watermarking for the Security of Teleradiology**", **IEEE** Sponsored World Conference on Futuristic Trends in Research and Innovation for Social Welfare (**WCFTR'16**), ISBN: 978-1-4673-9214-3, March 2016.

16. **V. Arulalan**, Dr. K. Suresh Joseph, "**Score Level Fusion of Iris and Finger Knuckle Print**", **IEEE** 10th International Conference on Intelligent Systems and Control (**ISCO'16**), ISBN: 978-1-4673-7807-9, January 2016.

17. G. Balamurugan, Sakthivel Punniakodi, K. Rajeswari, **V. Arulalan**, "**Automatic Number Plate Recognition System Using Super-Resolution Technique**", 2015 IEEE International Conference on Computing and Communications Technologies (**ICCCT'15**), ISBN:978-1-4799-7623-2/15/$31.00 ©2015 **IEEE**, February 2015.

18. G. Balamurugan, Dr. K. B. Jayarraman, **V. Arulalan**, "**An Iris Based Watermarking System for the Security Issues of Teleradiology**", International Journal of Applied Engineering Research, Vol. 9 No. 20, 2014. [**SCOPUS Indexed**]

IN PEER-REVIEWED NATIONAL CONFERENCES

19. **V. Arulalan,** DR. K. Suresh Joseph and G. Balamurugan, "**Safety Driver: Reducing Driver Distraction Using Integrated Smart Sensors**", Advances in Natural and Applied Sciences, Volume 11, Number 12, October 2017, Pages: 63-69, **ICMR** sponsored National

Conference on "ICT Solutions for Challenges and Issues in e-Health" (NCICTEH'17). [**SCOPUS Indexed**]

20. G. Balamurugan, Dr. K. B. Jayarraman, **V. Arulalan**, V. Lokesh, "**Multimodal Biometric Key Generation for Cryptographic Security using FACE and IRIS**", Advances in Natural and Applied Sciences, 9(6) Special 2015, Pages: 525-530, DSTE Sponsored National Conference on Intelligent Computing, January 2015 [**SCOPUS Indexed**]

D. RESEARCH METRICS – PROFILES

ORCID - https://orcid.org/0000-0001-9312-2143

GOOGLE SCHOLAR - http://scholar.google.co.in/citations?user=o-D6QfUAAAAJ

SCOPUS - https://www.scopus.com/authid/detail.uri?authorId=56809436800

WEB OF SCIENCE - https://www.webofscience.com/wos/author/record/3865260

DBLP - https://dblp.org/pid/324/6475.html

FDP's & WORKSHOP's ATTENDEND

1. Attended Two-Weeks Online FDP on "**Computer Vision and Natural Language Processing**" from 08-08-2022 to 18-08-2022 organized by Department of Information Technology, Easwari Engineering College, Chennai.
2. Attended two days online workshop on "**Effective paper writing and presentation with latex**" held during 23rd and 24th September 2022 organized by Department of Micro and Nano Electronics, Vellore Institute of Technology, Vellore.
3. Attended Five-days Online FDP on "**Research Trends and Methodologies in Machine Learning**" from 25-07-2022 to 29-07-2022 organized by Department of Computer Science & Engineering, Haldia Institute of Technology, West Bengal.
4. Attended Six-Days AICTE sponsored Short Term Training Programme on "**Recent Trends in Data Sciences**" from 03-01-2022 to 08-01-2022 organized by Department of Information Technology, Anna university, MIT Campus, Chennai.
5. Attended Five-days AICTE Training and Learning (ATAL) Academy Online FDP on "**Advanced Artificial Intelligence for Computer Vision**" from 26-07-2021 to 30-07-2021 at Indian Institute of Technology, Roorkee.

6. Attended Five-days AICTE Training and Learning (ATAL) Academy Online FDP on "**Artificial Intelligence for Computer Vision- Elementary**" from 12-07-2021 to 16-07-2021 at Indian Institute of Technology, Roorkee.
7. Attended Five-days AICTE Training and Learning (ATAL) Academy Online FDP on "**Quantum Computing**" from 21-06-2021 to 25-06-2021 at Birla Institute of Technology Mesra.
8. Attended Five-days AICTE Training and Learning (ATAL) Academy Online FDP on "**Creating Innovation & Startups with Internet of Things**" from 07-06-2021to 11-06-2021 at Institute of Technology & Management.
9. Attended three days Virtual International Workshop on "**Deep Learning for Computer Vision**" held during 08th-10th MAY 2021 organised by School of Computer Science and Engineering at Vellore institute of Technology, Chennai.
10. Attended Five-day AICTE Training and Learning (ATAL) Academy Online FDP on "**Electrical & Computer Engineering**" from 01-02-2021 to 05-02-2021 at National Institute of Technology, Andhra Pradesh.
11. Attended Five-day AICTE Training and Learning (ATAL) Academy Online FDP on "**AI Applications in Electrical & Computer Engineering**" from 04-01-2021 to 08-01-2021 at National Institute of Technology, Raipur.
12. Attended Five-day AICTE Training and Learning (ATAL) Academy Online FDP on "**Data Science**" from 15-12-2020 to 19-12-2020 at National Institute of Technology, Mizoram.
13. Attended One Week Online International Faculty Development Programme on **"Machine Learning and Computer Vision: Applications, Research Challenges (MLCV 2020)"** held during 24th-28th August 2020 organized as twinning programme by National Institute of Technology Silchar, Assam and Gauhati University Institute of Science & Technology, Guwahati, Assam.
14. Attended One Week Online International Faculty Development Programme on "**Python Programming**", held during 11-16 MAY 2020 organised by Geethanjali Institute of Science and Technology in association with Spoken Tutorial, IIT Bombay-PMMMNMTT.
15. Attended one-day online Faculty Development Training Programme on "**Machine Learning for Beginners**" held on 9th May 2020 organized by Department of Electrical and Electronics Engineering, Sathyabama Institute of Science and Technology, Chennai in association with Indian Servers.

16. Attended two days Research Oriented Faculty Development Training Programme (UGC-XII Plan) on **"Applied Machine Learning"** held during 09th-10th March 2020 in the Department of Instrumentation Engineering, MIT Campus, Anna University, Chennai.
17. Attended five days AICTE sponsored ATAL FDP on "**Artificial Intelligence**" held during 11-15 November 2019 in the Department of Information Technology, MIT Campus, Anna University, Chennai.
18. Attended one day workshop on "**Intellectual Property Rights-2019 (IPR-2019)**" held on 27th September 2019 at AC Tech Campus, Anna University, Chennai.
19. Attended two days workshop on "**Hands-on Training on Artificial Neural Network Using MATLAB**" held during 3rd August and 4th August at School of Electrical Engineering, Vellore Institute of Technology, Chennai.
20. Attended one day workshop on **"Global Azure Bootcamp"** held on 27th April 2019 at Tech Mahindra in association with **MICROSOFT INDIA**.
21. Attended two days IET Chennai Local Network sponsored Workshop on **"Machine Learning in Computer Vision using Python"** held during 27th and 28th March 2019 at SRM Institute of Science and Technology.
22. Attended a one-day workshop on **"Machine Learning od Industry 4.0"** held on 2nd March 2019 at the Department of Manufacturing Engineering, CEG Campus, Anna University.
23. Attended one day workshop on "**Transforming India with Deep Learning**" held on 9th October 2018, organized by NVIDIA Developer Connect, NVIDIA.
24. Attended three days Workshop on "**Image Processing, Computer Vision and Machine Learning**" held during 3rd – 5th MAY 2018 organized by the Department of Electronics Engineering at Madras Institute of Technology, Anna University.
25. Attended one day workshop on "**How to Do a Good Ph.D.**" held on 3rd March organized by Centre for Aerospace Research at Madras Institute of Technology, Anna University.
26. Attended one day workshop on "**URKUND (Plagiarism Detection Tool)**" held on 28th October 2015 at Pondicherry University, Organized by Ananda Rangapillai Library, Pondicherry University in Collaboration with INFLIBNET Centre, Gujarat.
27. Attended two days CSI sponsored Workshop on "**Image processing using MATLAB**" held during 23rd and 24th February 2015 at Manakula Vinayagar Institute of Technology, Puducherry.
28. Attended five days national workshop on "**Advanced Trends in Optimization Methods & their Applications (ATOMA-2014)**" held from 8th September to 12th September 2014 at Pondicherry University, Puducherry.

29. Attended national workshop on "**Image Processing Techniques using MATLAB**" held during 5th and 6th September 2014 at Kongu Engineering College, Erode.
30. Attended five days national workshop on "**Research Trends in Digital Image Processing NWRTDIP 2014**" held from 18th August to 22nd August 2014 at Pondicherry University, Puducherry.
31. Attended two days workshop on "**Big Data & Open Cloud**" held during 3rd April and 4th April 2014 at Pondicherry University, Puducherry.
32. Attended a one-day national level workshop on "**Mobile Applications Development Using Android: A Practical Approach (WMADA 2014)**" held during 15th March 2014 at Pondicherry University, Puducherry.
33. Attended "**Author Workshop**" jointly organized by **Springer and Ananda Rangapillai Library** held on 6th February 2014 at Pondicherry University, Puducherry.
34. Attended AICTE Sponsored two days national seminar on "**Impact of Biometric Security Using Image Processing"** held during 12th November and 13th November 2013 at Manakula Vinayagar Institute of Technology, Puducherry.

PROFESSIONAL SOCIETY MEMBERSHIP

1. Professional Member in Association for Computing Machinery (**ACM**)
   MEMBERSHIP NO: **9451841**.

2. Senior Member in International Society for Research and Development (**ISRD**)
   MEMBERSHIP NO: **SR3140900615**

3. Life Member in International Association for The Engineers (**IAENG**)
   MEMBERSHIP NO: **172053**.

4. Student Member [UACEE] in Institute of Research Engineers and Doctors (**IRED**)
   MEMBERSHIP NO: **SM10100059181**

EXTRACURRICULAR ACTIVITIES

1. An Active member in **Puduvai GLUG** (GNU/Linux User Group), Puducherry.

2. An Active member in **PyCon INDIA**.

PERSONAL PROFILE

Date of Birth	27-02-1991
Father's Name	Dr. L. Veerappan
Nationality	Indian
Gender	Male
Linguistic Proficiency	English, Tamil, Hindi
Residence Address	S2, D Square Raj Paradise, Chitlapakkam, Chennai-600064.
Permanent Address	31, New Street, Pillaichavady, Puducherry-605014.

REFERENCES

Dr. Dhananjay Kumar M.E., M.Tech., Ph.D.,
Professor and Head,
Department of Information Technology,
Madras Institute Of Technology Campus,
Anna University, Chennai - 600044.
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Dr. K. Suresh Joseph M.E., Ph.D.,
Associate Professor,
Department of Computer Science,
School of Engineering and Technology,
Pondicherry University,
Puducherry - 605014.
Mobile: +91-9444321492
Email: ksjoseph.csc@pondiuni.edu.in
ksjoseph.csc@gmail.com

DECLARATION

I hereby solemnly declare that all information furnished here are true to the best of my knowledge and conscience.

PLACE : Chennai



(V. ARULALAN)